

Online Bookstore Website – Testing Document

1. Introduction

This testing document defines the test scenarios and detailed test cases for the Online Bookstore Web Application. It ensures that the application meets the defined functional and non-functional requirements. The tests will serve as a foundation for future testing activities as development progresses.

2. Test Scenarios

1. Book Browsing and Search
 - Verify users can browse books by category.
 - Verify search functionality by title, author, or category.
2. Shopping Cart
 - Verify users can add, remove, and update book quantities in the cart.
 - Verify total price updates dynamically.
3. Checkout Process
 - Verify cart details are correctly displayed during checkout.
 - Verify users can input shipping details and apply discount codes.
4. Payment Processing
 - Verify payment details are validated.
 - Verify successful/failed payments show correct messages.
5. Order Confirmation
 - Verify order confirmation page and email are generated.
6. User Account Management
 - Verify account creation, login, logout, and profile update.
 - Verify users can view order history.
7. Responsive Design

- Verify site adapts to desktop, tablet, and mobile views.

8. Performance & Security (Non-Functional)

- Verify site loads quickly and supports 1000 concurrent users.
- Verify sensitive data is encrypted and app resists common vulnerabilities.

3. Test Cases

3.1 Book Browsing and Search Functionality

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC001-01	Verify browsing by category	Homepage accessible	Navigate to /books?category=Fiction	Category = Fiction	List of Fiction books displayed (title, price, description)	Fiction books displayed in results	High
TC001-02	Verify browsing by unknown category	Homepage accessible	Navigate to /books?category=Unknown	Category = Unknown	Either 404 error OR "no results" message displayed	Correct handling of unknown categories	Medium
TC001-03	Verify search by keyword	Search bar visible	Enter keyword in /search?q=Python	"Python"	Matching books displayed or "no results" message shown	Search results relevant or empty state displayed	High

TC001-04	Verify search is case insensitive and trims whitespace	Search bar visible	Perform two searches: /search?q=Python and /search?q=%20pYtHon%20	"Python", "pYtHon"	Both queries return the same (or equivalent) results	Results consistent regardless of case/whitespace	Medium
TC001-05	Verify empty search query handling	Search bar visible	Perform search /search?q=	Empty query	Returns error (400) OR a friendly message ("Enter search term")		

3.2 Shopping Cart Functionality

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC002-01	Verify adding book to cart	Book listing visible	Click "Add to Cart" for a book	Book A	Book added to cart, cart shows item + price	Book appears in cart	High
TC002-02	Verify updating quantity	Cart has items	Increase quantity to 2	Qty = 2	Total updates correctly	Total reflects new qty	High
TC002-03	Verify removing book	Cart has items	Click "Remove"	N/A	Item removed, total updated	Cart updates correctly	High

3.3 Checkout Process

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC003-01	Verify checkout page displays details	Cart has items	Go to checkout page	N/A	Cart details shown (items, qty, total)	Correct info displayed	High
TC003-02	Verify entering shipping info	On checkout page	Fill in address form	Name, Addresses	Info saved and confirmed	Valid info accepted	High
TC003-03	Verify applying discount code	On checkout page	Enter discount code	"SAVE10"	Discount applied	Total reflects discount	Medium

3.4 Payment Processing

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC004-01	Verify valid payment	On checkout page	Enter valid credit card	Card 4111111111111111	Payment successful → Redirect to confirmation	Confirmation page displayed	High
TC004-02	Verify failed payment	On checkout page	Enter expired card	Card 4000000000000002	Payment fails	Error message shown	High

3.5 Order Confirmation

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC005-01	Verify confirmation page	Successful payment	Complete checkout	Valid order	Order summary displayed	Correct order info shown	High
TC005-02	Verify email confirmation	Successful payment	Complete checkout	Valid email	Email received	Email matches order details	High

3.6 User Account Management

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC006-01	Verify account creation	Registration page accessible	Fill email & password → Submit	user@test.com	Account created	User redirected to profile	High
TC006-02	Verify login/logout	Account exists	Enter valid credentials → Logout	Email + Password	Login succeeds → Logout works	User logged out	High
TC006-03	Verify view past orders	Logged-in user with history	Go to order history	N/A	Past orders displayed	Correct history shown	Medium

3.7 Responsive Design

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC007-01	Verify mobile responsiveness	Mobile device	Open site	N/A	Layout adapts to screen	Mobile layout displayed	Medium

TC007-02	Verify tablet responsiveness	Tablet device	Open site	N/A	Layout adapts to screen	Tablet layout displayed	Medium
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3.8 Non-Functional Testing

Test Case ID	Objective	Pre-Conditions	Test Steps	Input Data	Expected Result	Pass/Fail Criteria	Priority
TC008-01	Verify performance load	Load test tool	Simulate 1000 users	1000 concurrent requests	System stable	Response < 3s	High
TC008-02	Verify SSL/TLS encryption	Site deployed	Open checkout page	N/A	HTTPS with valid SSL	Secure connection enforced	High
TC008-03	Verify SQL Injection protection	Search bar	Enter ' OR 1=1 --	Malicious input	Input sanitized	No vulnerability	High